

Electronic Supplementary Material:
Support Flow Tensor Field for Fast Build-Orientation Candidate
Generation

Supplementary Tables

Table S1: Computation time of the four implementations for the 35 meshes of the five groups. t_{CPU} and t_{CUDA} are the TOMO_CPU/TOMO_CUDA 1° exhaustive sweep ($\theta_c = 60^\circ$) DLL times respectively, t_{py} is the Python SFTF candidate evaluation, and $t_{\text{C++}}$ is the SFTF_C++ pure computation time. hit is the number of valid face-to-face supports of the Python SFTF top direction. The speed columns are $t_{\text{CPU}}/t_{\text{py}}$ and $t_{\text{CPU}}/t_{\text{C++}}$ respectively.

ID	Shape	Faces	hit	$t_{\text{CPU}}(\text{s})$	$t_{\text{CUDA}}(\text{s})$	$t_{\text{py}}(\text{s})$	$t_{\text{C++}}(\text{ms})$	CPU/py	CPU/C++
<i>A: basic shapes</i>									
A1	cube	12	0	49.64	11.45	0.13	4.1	382×	12,108×
A2	sphere	2,976	498	40.22	13.82	0.74	23.9	54×	1,683×
A3	cylinder	192	0	39.64	22.96	0.17	1.3	233×	30,494×
A4	cone	96	0	40.51	14.99	0.12	0.8	338×	50,641×
A5	torus	2,304	184	43.72	13.57	0.45	14.2	97×	3,079×
<i>B: simple parts</i>									
B1	u_bracket	28	2	40.34	13.91	0.15	3.9	269×	10,344×
B2	hook	1,560	235	41.48	12.51	0.58	15.8	72×	2,626×
B3	c_clamp	28	1	44.82	17.43	0.17	0.8	264×	56,027×
B4	pipe_elbow	2,400	500	46.99	13.39	0.81	25.3	58×	1,857×
B5	hollow_box	24	4	38.63	17.47	0.21	0.9	184×	42,921×
<i>C: organic benchmarks</i>									
C1	<i>Bunny 69k</i>	69,662	81	47.46	52.55	1.69	247.7	28×	192×
C2	<i>Manikin 13k</i>	13,672	134	37.85	30.44	1.02	44.3	37×	854×
C3	<i>Dragon 100k</i>	99,999	970	73.81	94.23	3.17	323.3	23×	228×
C4	<i>Happy 50k</i>	49,944	240	47.15	51.71	1.50	130.1	31×	363×
C5	<i>Lucy 50k</i>	49,999	542	51.15	71.73	1.87	172.5	27×	297×
<i>D: Thingi10K set 1</i>									
D1	37009	23,909	57	51.46	144.10	1.37	114.5	38×	449×
D2	37010	13,247	35	45.58	148.94	1.24	63.5	37×	718×
D3	37011	17,288	82	52.65	59.75	1.15	89.2	46×	590×
D4	37091	256	42	46.38	21.23	0.36	5.2	129×	8,920×
D5	37095	468	65	38.19	39.54	0.49	6.2	78×	6,160×
D6	37111	88,106	386	99.34	212.38	3.42	465.4	29×	213×
D7	37137	64,078	434	73.92	233.45	2.90	305.5	25×	242×
D8	37402	8,322	119	40.32	73.76	2.43	49.7	17×	811×
D9	37415	13,340	77	44.51	126.27	2.96	65.4	15×	681×
D10	37416	12,036	496	43.45	64.44	3.39	80.4	13×	540×
<i>E: Thingi10K set 2, large meshes</i>									
E1	39507	969,520	562	973.29	1041.71	16.92	4387.6	58×	222×
E2	45809	517,589	107	513.22	585.85	9.71	1984.5	53×	259×
E3	45811	1,037,876	1,309	1038.40	1153.79	18.69	4515.6	56×	230×
E4	46012	638,754	50	658.57	802.44	11.15	2583.8	59×	255×
E5	55280	900,490	1,072	939.61	1102.83	16.78	3957.9	56×	237×
E6	59197	786,432	699	776.49	826.30	17.50	4119.5	44×	188×
E7	61384	516,608	1,766	521.54	642.22	12.85	2065.0	41×	253×
E8	64445	514,938	2,288	125.46	211.96	15.05	2208.0	8.3×	57×
E9	64446	514,938	2,288	125.66	211.60	14.93	2211.4	8.4×	57×
E10	65942	754,688	1,175	765.24	830.14	22.11	3201.6	35×	239×

Table S2: Cost decomposition of the SFTF top direction of representative meshes. Convex basic shapes (cube, cone) are pure build-plate support with $\text{hit} = 0$, $R = P = 0$, and the rest show face-to-face self-support (P) and build-plate support (B) together.

ID	Shape	hit	R	P	B
A1	cube	0	0	0	401.5
A4	cone	0	0	0	179.8
A2	sphere	498	47.3	109.5	0.02
A5	torus	184	15.9	40.3	209.1
B2	hook	235	19.9	55.4	128.7
C1	<i>Bunny 69k</i>	81	1.4	25.2	176.3
C3	<i>Dragon 100k</i>	970	109.4	267.6	417.5
D6	37111	386	227.9	932.8	1,613
D10	37416	496	6,381	12,190	1.3

Table S3: TOMO_CPU global minimum v_{ss} by critical angle (representative meshes, 1° grid). The support volume decreases monotonically as θ_c increases. In the region of large θ_c , almost all faces become self-supporting and support is essentially unnecessary, in which case the numerical residual of the MSST decomposition $V_{ss} = V_{tc} - V_o - V_{nv}$ can make v_{ss} appear as a small negative number (effectively 0).

ID	Mesh	$\theta_c=0^\circ$	30°	45°	60°	90°
A1	cube	1,750	1,734	0	0	0
B2	hook	1,736	713	319	128	0
C1	<i>Bunny 69k</i>	49,031	23,664	13,970	3,201	-11,049
C3	<i>Dragon 100k</i>	214,783	117,195	82,554	24,447	-18,087
D1	37009	65,814	4,121	461	-1,238	-10,002
D6	37111	99,197	54,214	36,036	9,703	-10,653
E1	39507	26,341	9,969	2,262	-10,252	-19,031
E7	61384	72,428	19,331	12,528	-3,847	-18,559

Table S4: Oracle (best within candidate pool) ratio by the number of coarse directions. Meshes other than *Happy 50k* already include a near-optimal direction at 512.

# coarse directions	<i>Bunny 69k</i>	<i>Manikin 13k</i>	<i>Dragon 100k</i>	<i>Happy 50k</i>	<i>Lucy 50k</i>
512	1.46	1.13	1.29	4.35	1.02
1024	1.50	1.11	1.33	3.94	1.02
2048	1.61	1.06	1.28	1.91	1.04
4096	1.62	1.13	1.27	1.91	1.30

Table S5: Best-of-3 ratio of the objective-function ablation (lower is better). The proposed objective function $J = R + P + B$ and the old objective function including nuclear/ σ_1 are completely identical per mesh.

Ranking variant	<i>Bunny 69k</i>	<i>Manikin 13k</i>	<i>Dragon 100k</i>	<i>Happy 50k</i>	<i>Lucy 50k</i>	Average
T (rank tuned, in-sample)	1.46	1.86	1.62	11.23	2.82	3.80
$J = R + P + B$ (proposed)	2.47	3.99	2.01	11.15	2.82	4.49
$R+P+B+0.05S+0.05\sigma_1$ (old objective)	2.47	3.99	2.01	11.15	2.82	4.49
B alone	2.57	2.60	1.94	16.04	2.82	5.19
$0.05S + 0.05\sigma_1$ (tensor alone)	5.82	5.38	6.57	11.86	5.62	7.05
P alone	5.82	4.92	6.27	12.07	7.51	7.32

Table S6: Best-of-3 ratio when sorting candidates by $J = R + P + B$ and the unified term $\tilde{R} = R+B$ (lower is better). On the average excluding *Happy 50k* (a narrow-basin hard case under the ranking-only point metric, resolved by local verification in the main text), $\tilde{R}$ outperforms J .

Mesh	$J = R+P+B$	$\tilde{R} = R+B$
<i>Bunny 69k</i>	2.47	2.57
<i>Manikin 13k</i>	3.99	2.89
<i>Dragon 100k</i>	2.01	1.70
<i>Lucy 50k</i>	2.82	2.82
Average (excl. <i>Happy 50k</i>)	2.82	2.50

Supplementary Figures

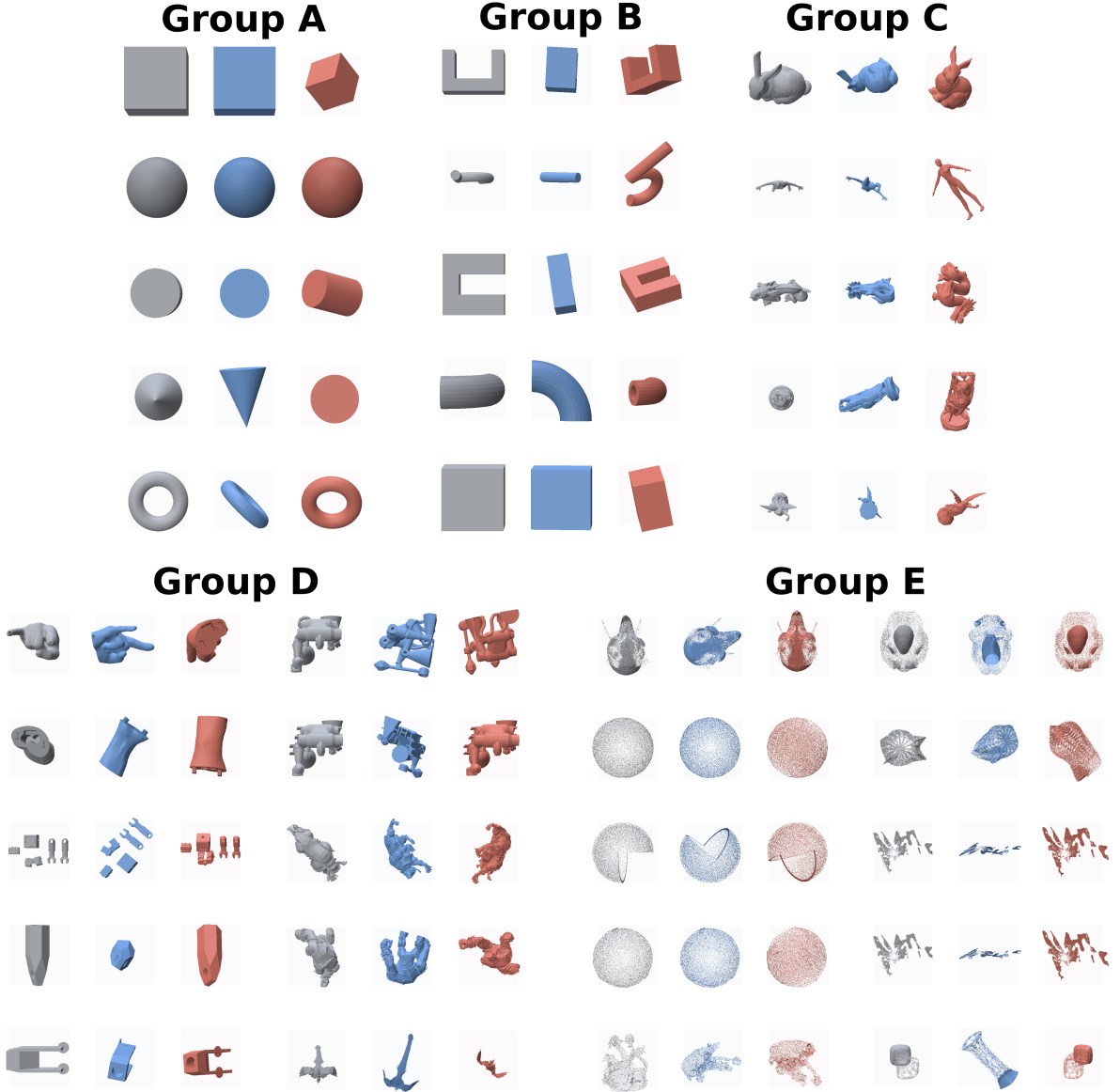


Figure S1: Best/worst build-orientation Polyscope overlay arranged by group for the five groups (A–E) ($\theta_c = 60^\circ$). A label is placed atop each group, with groups A–C in the top row and groups D/E (a vertical 5-row 2-column layout each) in the bottom row, grouped about the page center. Each mesh shows the original (gray), optimal orientation (blue), and worst orientation (red) side by side, displayed in the actual print pose by rotating the stored build direction to world $+Z$. Since SFTF candidates are invariant to θ_c , the overlays of the five critical angles are identical.

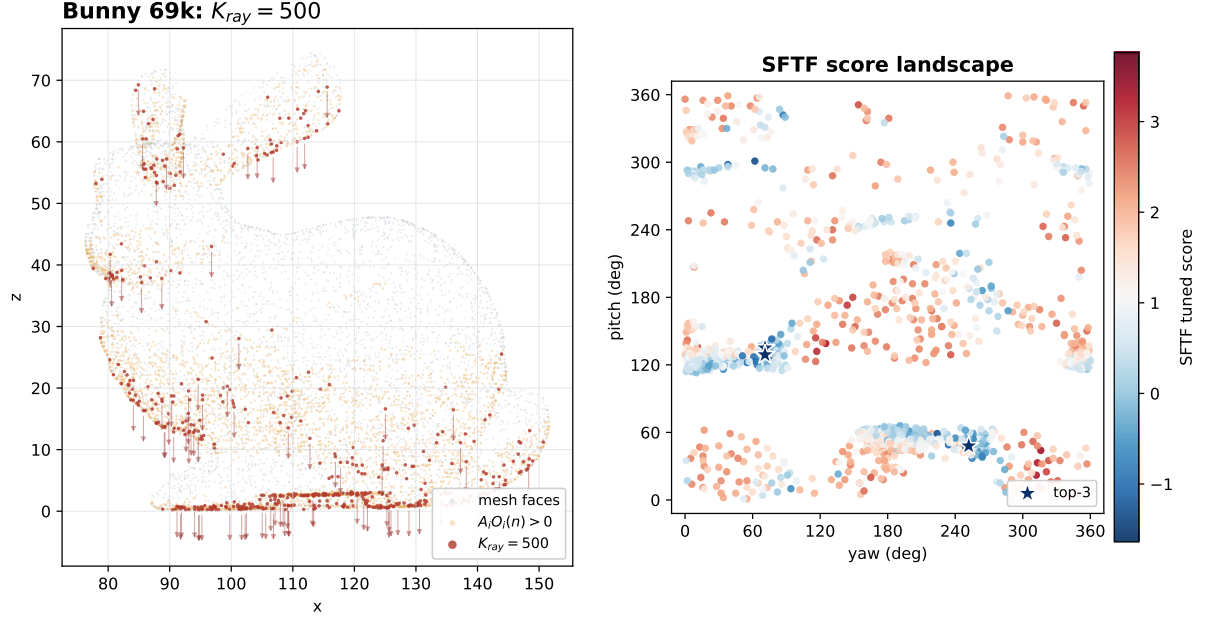


Figure S2: *Bunny 69k* source-face sampling and SFTF score landscape with fixed $K_{ray} = 500$. The right panel is recomputed with the same ray budget and uses the common color scale of Supplementary Figs. S3–S4.

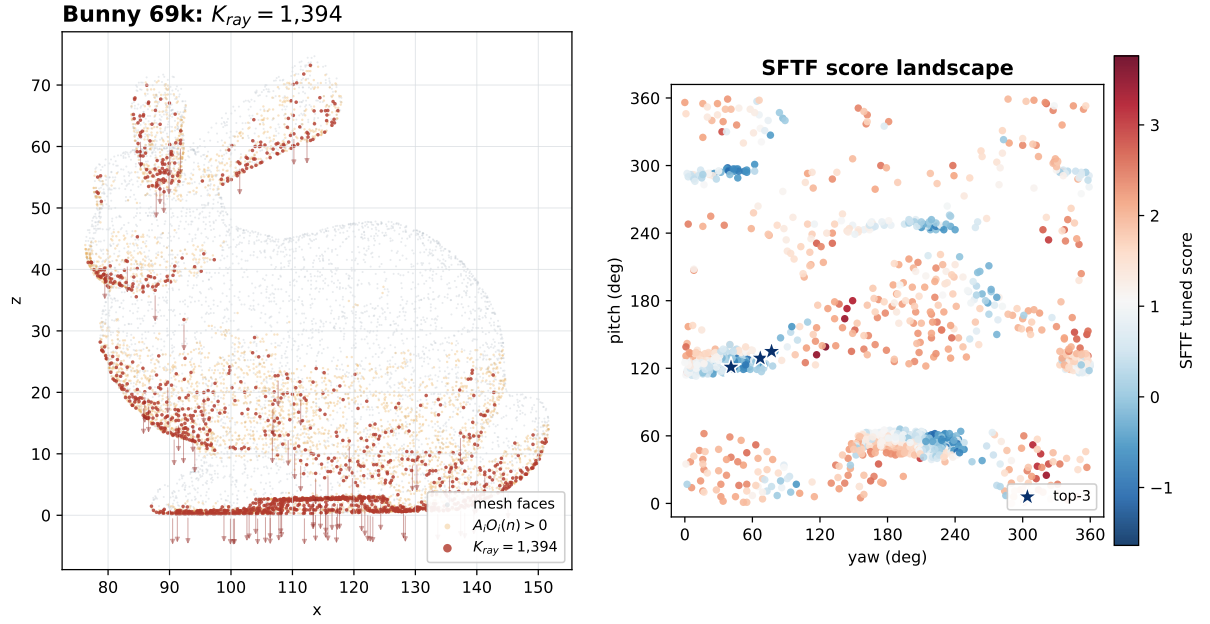


Figure S3: *Bunny 69k* source-face sampling and SFTF score landscape with the adaptive rule, $K_{ray} = \text{clip}(0.02N_f, 500, 3000) = 1,394$.

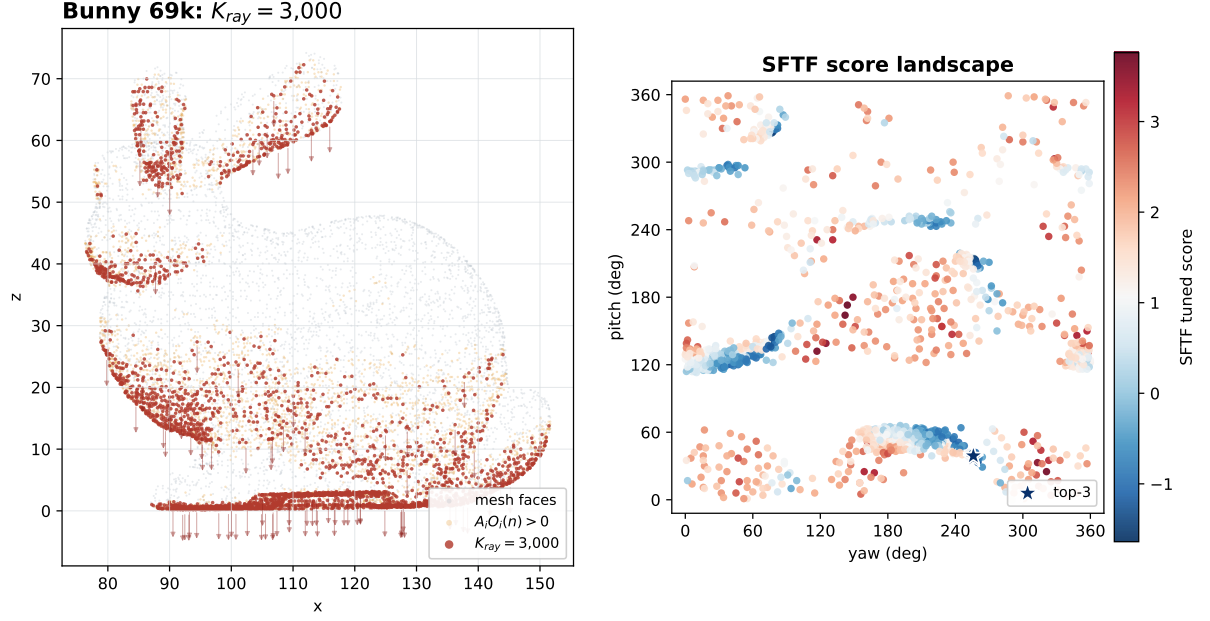


Figure S4: *Bunny 69k* source-face sampling and SFTF score landscape with fixed $K_{ray} = 3000$, corresponding to the upper cap used for large meshes.

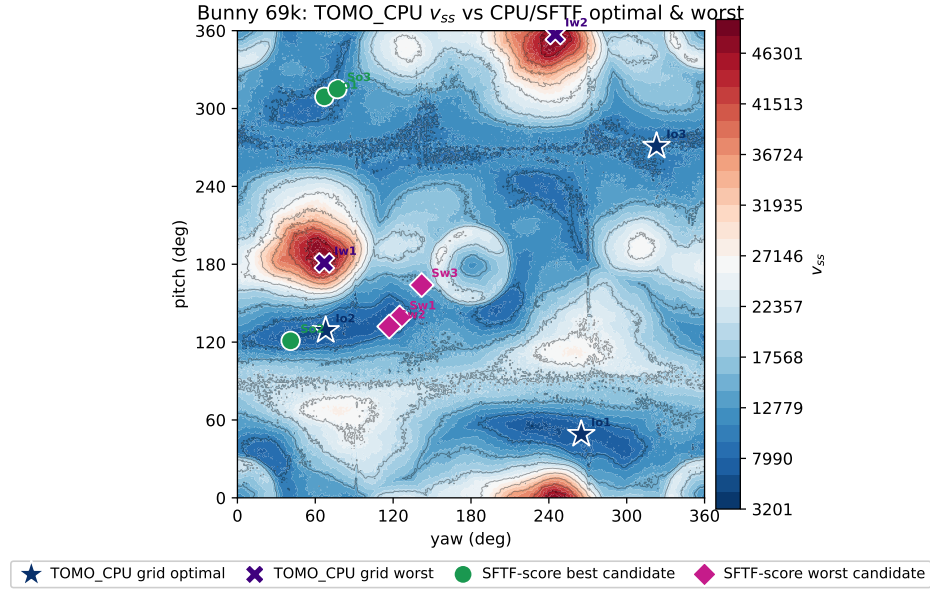


Figure S5: Comparison of CPU/SFTF optimal/worst orientations on the TOMO_CPU v_{ss} landscape of *Bunny 69k* (blue is good). The markers are 3 each of CPU/SFTF optimal/worst, with kinds following the legend below the figure. The global optimum is $v_{ss} = 3201$ (yaw= 265° , pitch= 49° ; the main-text comparison table), and lies not at a single point but within a broad low-cost basin.

